

Problem 1.3

Solution:**(a)**

Viewing A as n/ℓ row blocks of size $\ell \times \ell$ and B as n/ℓ column blocks of size $\ell \times \ell$, each multiplication involves two $\ell \times \ell$ matrix multiplication, Strassen's algorithm computes the product of two $m \times m$ matrices in $O(m^{2.81})$ time, so the total complexity is $O(\ell^{2.81}(n/\ell)^2) = O(\ell^{0.81}n^2)$.

(b)

For each symbol $a \in \Sigma$, define the following two $n \times \ell$ $\{0, 1\}$ -matrices:

$$X_a(i, k) = \begin{cases} 1, & \text{if } s_k^{(i)} = a, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$Z_a(i, k) = \begin{cases} 1, & \text{if } s_k^{(i)} \in \Sigma \setminus \{a\}, \\ 0, & \text{if } s_k^{(i)} = a \text{ or } s_k^{(i)} = ?. \end{cases}$$

The idea is that if at some position k we have $s_k^{(i)} = a$ and $s_k^{(j)} \neq a$ (and in particular, $s_k^{(j)} \in \Sigma$, not $?$) then the pair (i, j) cannot match.

For each $a \in \Sigma$, compute the $n \times n$ matrix

$$M_a = X_a \cdot (Z_a)^T,$$

where entry (i, j) in M_a counts the number of positions k such that $s_k^{(i)} = a$ and $s_k^{(j)} \in \Sigma \setminus \{a\}$.

Then, define

$$M(i, j) = \sum_{a \in \Sigma} M_a(i, j).$$

Notice that if $M(i, j) > 0$, then there exists at least one position where $s^{(i)}$ and $s^{(j)}$ conflict (i.e., they both have a letter from Σ but the letters differ). Therefore, if $M(i, j) = 0$ for some $i \neq j$, then for every position k either $s_k^{(i)} = s_k^{(j)}$, or at least one of $s_k^{(i)}$ and $s_k^{(j)}$ is $?$; that is, $s^{(i)}$ and $s^{(j)}$ match.

Each multiplication $X_a \cdot (Z_a)^T$ is the product of an $n \times \ell$ and an $\ell \times n$ matrix. By part (a), each such product can be computed in $O(\ell^{0.81}n^2)$ time. Since there are only $|\Sigma|$ symbols (and typically $|\Sigma|$ is a constant), the total time is $O(\ell^{0.81}n^2)$.